

Task-4: Auto Scaling & Load Balancer

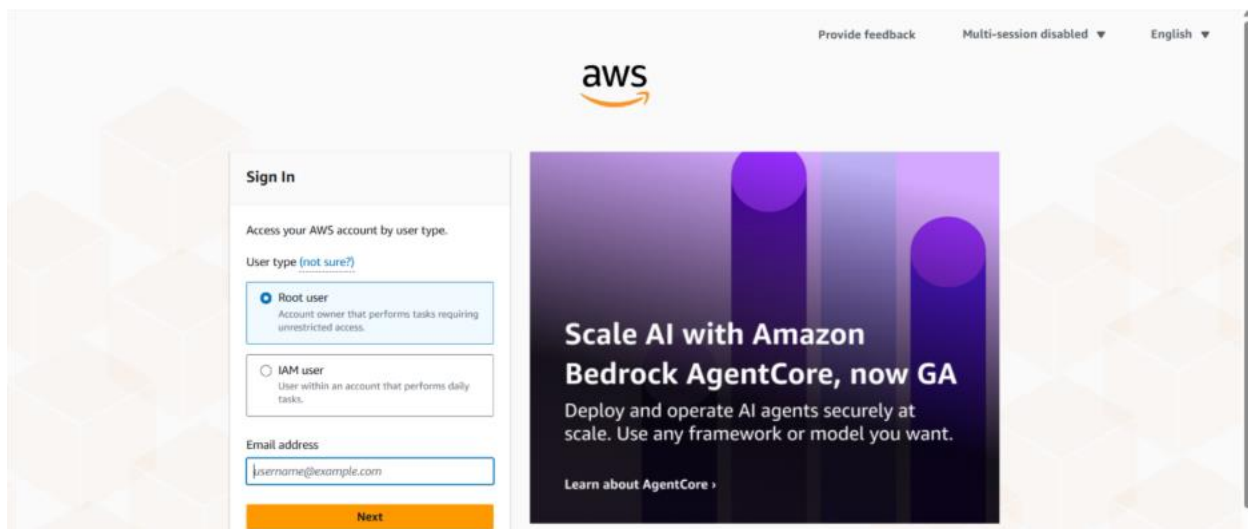
1. Create two private EC2 instances in different Availability Zones.
2. Configure Apache (httpd) on the instances.
3. Create an Application Load Balancer (ALB).
- 4. Create an Auto Scaling Group and attach it to the Load Balancer.**
5. Verify that:
 - a. The website is accessible using the Load Balancer DNS.
 - b. The application is still accessible even if one EC2 instance is stopped.

Solution:

1. Login to your AWS account with credentials and required verification.

(<https://signin.aws.amazon.com/>)

My account – AWS Management Console- Sign in



2. Create a custom VPC:
 - Goto – VPC- Create VPC- Select VPC and More
 - Name – my-vpc
 - Assign IPV4 CIDR Block – 10.0.0.0/16
 - (CIDR block must be in between /16 to /24)
 - Tendency – Default
 - This will create 2 public subnet, 2 private Subnet, NAT Gateway and an Internet Gateway.

VPC > Your VPCs > Create VPC

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
Task4-

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IP's
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)
Default

► **Encryption settings - optional**

Preview

Subnets (4)
Subnets within this VPC

us-east-1a

- Task4--subnet-public1-us-east-1a
- Task4--subnet-private1-us-east-1a

us-east-1b

- Task4--subnet-public2-us-east-1b
- Task4--subnet-private2-us-east-1b

Route tables (3)
Route network traffic to resources

- Task4--rtb-public
- Task4--rtb-private1-us-east-1a
- Task4--rtb-private2-us-east-1b

Network connections (3)
Connections to other resources

- Task4--igw
- Task4--nat-public1
- Task4--nat-public2

Chose Subnets

VPC > Your VPCs > Create VPC

Tenancy [Info](#)
Default

► **Encryption settings - optional**

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 2 3

► **Customize AZs**

Number of public subnets [Info](#)
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the Internet.

0 2 4

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 2 4

► **Customize subnets CIDR blocks**

NAT gateways (\$) - updated [Info](#)

Preview

Subnets (4)
Subnets within this VPC

us-east-1a

- Task4--subnet-public1-us-east-1a
- Task4--subnet-private1-us-east-1a

us-east-1b

- Task4--subnet-public2-us-east-1b
- Task4--subnet-private2-us-east-1b

Route tables (3)
Route network traffic to resources

- Task4--rtb-public
- Task4--rtb-private1-us-east-1a
- Task4--rtb-private2-us-east-1b

Network connections (3)
Connections to other resources

- Task4--igw
- Task4--nat-public1
- Task4--nat-public2

Chose NAT Gateway and make sure each AZ have seprate NAT Gateway.

VPC > Your VPCs > Create VPC

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 2 4

► **Customize subnets CIDR blocks**

NAT gateways (\$) - updated [Info](#)
NAT gateway allows private resources to access the Internet from any availability zone within a VPC, providing a single managed Internet exit point for the entire region. Additional charges apply.

None | Regional - new | Zonal

NAT gateways (\$) [Info](#)
Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway.

In 1 AZ 1 per AZ

VPC endpoints [Info](#)
Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None | S3 Gateway

DNS options [Info](#)
☒ Enable DNS hostnames
☒ Enable DNS resolution

Preview

Subnets (4)
Subnets within this VPC

us-east-1a

- Task4--subnet-public1-us-east-1a
- Task4--subnet-private1-us-east-1a

us-east-1b

- Task4--subnet-public2-us-east-1b
- Task4--subnet-private2-us-east-1b

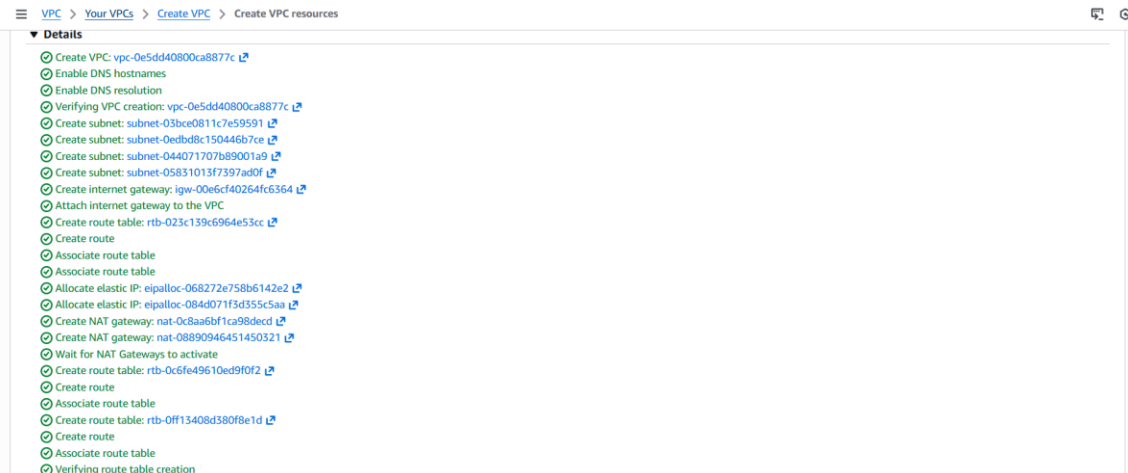
Route tables (3)
Route network traffic to resources

- Task4--rtb-public
- Task4--rtb-private1-us-east-1a
- Task4--rtb-private2-us-east-1b

Network connections (3)
Connections to other resources

- Task4--igw
- Task4--nat-public1
- Task4--nat-public2

Wait till all process of VPC creation is done



3. Create Security Groups:

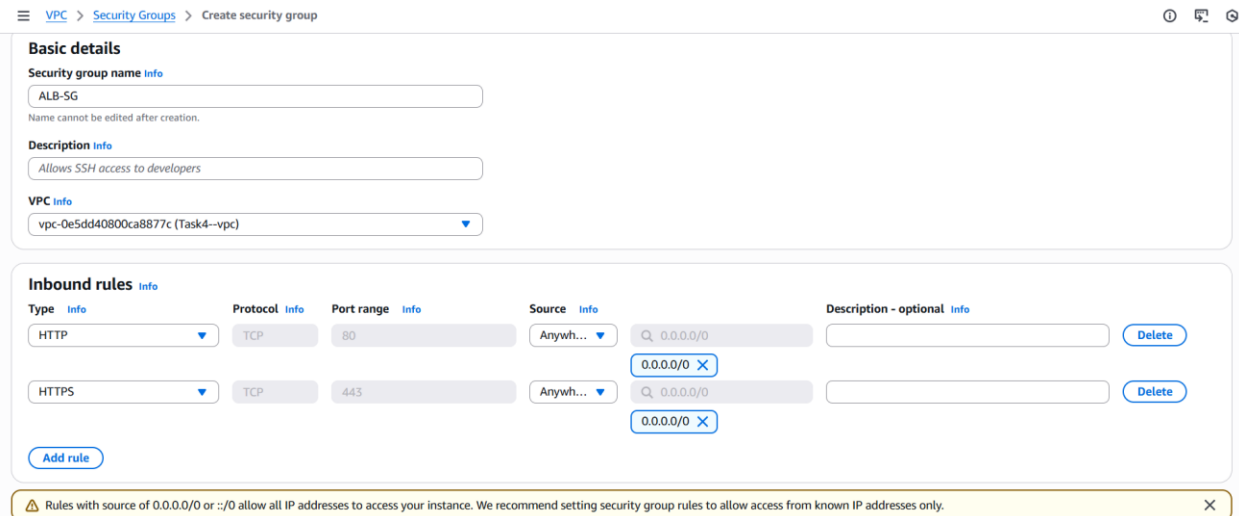
Application Load Balancer Security Group

EC2 → Security Groups → Create

Name- ALB-SG

Inbound – HTTP-80-0.0.0.0/0

Outbound – All traffic



EC2 Security Group

EC2 → Security Groups → Create

Name- EC2-SG

Inbound – HTTP-80-source: ALB-SG

SSH-22-Source: Your IP

Outbound – All traffic

Security group name Info
EC2-SG
Name cannot be edited after creation.

Description Info
EC2 Security Group

VPC Info
vpc-0e5dd4080ca8877c (Task4--vpc)

Inbound rules Info

Type	Protocol	Port range	Source	Description - optional	
HTTP	TCP	80	Custom sg-0a18bb2bae38e9d08		Delete
HTTPS	TCP	443	Custom sg-0a18bb2bae38e9d08		Delete
SSH	TCP	22	My IP sg-0a18bb2bae38e9d08 111.125.244.2/32		Delete

Add rule

4. Launch Private EC2 Instance in ec2 private subnet

EC2- Launch Instance-

- Name- Privateec2-1
- OS- Amazon Linux
- T2-micro
- Select Key Pair
- Select Existing Security Group- EC2-SG

Network settings Info

VPC - required Info
vpc-05d8c76cbaeb9e259 (my-vpc)

Subnet Info
subnet-032e9fcc92210ef74 (pv-subnet)

Auto-assign public IP Info
Disable

Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups Info
Select security groups
private-ec2-sg sg-00705905d188cca0d

Summary

Number of instances Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-068c0051b15c8b816

Virtual server type (instance type)
t3.micro

Firewall (security group)
private-ec2-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

EC2- Launch Instance-

- Name- Privateec2-2
- OS- Amazon Linux
- T2-micro
- Select Key Pair
- Select Existing Security Group- EC2-SG

EC2 > Instances > Launch an instance

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-05d8c76c9aeb9e259 (my-vpc)
10.0.0.0/16

Subnet [Info](#)

subnet-032e9fcc92210ef74 pv-subnet
VPC: vpc-05d8c76c9aeb9e259 Owner: 798974631771
Availability Zone: us-east-1a (use1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.2.0/24

[Create new subnet](#)

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

private-ec2-sg sg-00705905d188cca0d [X](#)
VPC: vpc-02486129a4c880ca

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-068c0051b15c8b816

Virtual server type (instance type)
t3.micro

Firewall (security group)
private-ec2-sg

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

5. Connect These instances using public server

Private server 1

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri Dec 19 09:41:47 2025 from 19.206.107.29
[ec2-user@ip-10-0-21-144 ~]$ ls
key.pem
[ec2-user@ip-10-0-21-144 ~]$ ssh -i "key.pem" ec2-user@10.0.128.113
The authenticity of host '10.0.128.113 (10.0.128.113)' can't be established.
ED25519 key fingerprint is SHA256:JES+1l9WkhuRTtAFmm7QwvnaNO0dRlF7uxVjPW+2Pmg.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.128.113' (ED25519) to the list of known hosts.

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-128-113 ~]$ sudo vi apache.sh
[ec2-user@ip-10-0-128-113 ~]$ ls
apache.sh
[ec2-user@ip-10-0-128-113 ~]$ sudo chmod +x apache.sh
[ec2-user@ip-10-0-128-113 ~]$ ./apache.sh
bash: ./ is a directory
[ec2-user@ip-10-0-128-113 ~]$ ./apache.sh
Amazon Linux 2023 kernel livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Fri Dec 19 10:10:50 2025.
Dependencies resolved.

Package Architecture Version Repository Size
i-0801c010326d9a8b0 (public2)
Sub-Info: 10.111.122.122 Sub-Info: 10.0.11.144
```

Private server 2

```
aws
[Alt+S]
United States (N. Virginia)
Account ID: 8896-1568-6095
Ajit Jadhav

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri Dec 19 09:41:47 2025 from 10.206.107.29
[ec2-user@ip-10-0-21-144 ~]$ ls
key.pem
[ec2-user@ip-10-0-21-144 ~]$ ssh -i "key.pem" ec2-user@10.0.128.113
The authenticity of host '10.0.128.113 (10.0.128.113)' can't be established.
ED25519 key fingerprint is SHA256:2f2uilll0Hhah0RPApaw7wvna00dMf7uaxjW*Jrug.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.128.113' (ED25519) to the list of known hosts.

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-128-113 ~]$ sudo vi apache.sh
[ec2-user@ip-10-0-128-113 ~]$ ls
apache.sh
[ec2-user@ip-10-0-128-113 ~]$ sudo chmod +x apache.sh
[ec2-user@ip-10-0-128-113 ~]$ ./apache.sh
-bash: ./: is a directory
[ec2-user@ip-10-0-128-113 ~]$ ./apache.sh
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:02 ago on Fri Dec 19 10:10:50 2025.
Dependencies resolved.

Package Architecture Version Repository Size
i-0801c010326d9a8b0 (public2)
B:lib64: 10.55.52c 52c B:lib64: 10.0.31.144
```

Make sure you install Apache server on both private instance

```
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.65-1.amzn2023.0.2.x86_64 13/13
Running scriptlet: httpd-2.4.65-1.amzn2023.0.2.x86_64 1/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
Verifying : apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64 3/13
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.65-1.amzn2023.0.2.x86_64 6/13
Verifying : httpd-core-2.4.65-1.amzn2023.0.2.x86_64 7/13
Verifying : httpdfilesystem-2.4.65-1.amzn2023.0.2.noarch 8/13
Verifying : httpd-tools-2.4.65-1.amzn2023.0.2.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13
Verifying : mod_lua-2.4.65-1.amzn2023.0.2.x86_64 13/13

Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64 apr-util-1.6.3-1.amzn2023.0.2.x86_64 apr-util-ldb-1.6.3-1.amzn2023.0.2.x86_64 apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.65-1.amzn2023.0.2.x86_64 httpd-core-2.4.65-1.amzn2023.0.2.x86_64 httpdfilesystem-2.4.65-1.amzn2023.0.2.noarch httpd-tools-2.4.65-1.amzn2023.0.2.x86_64 libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch mod_http2-2.0.27-1.amzn2023.0.3.x86_64 mod_lua-2.4.65-1.amzn2023.0.2.x86_64

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service -> /usr/lib/systemd/system/httpd.service.
* httpd.service - The Apache HTTP Server
Loaded loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
Active: active (running) since Fri 2025-12-19 10:10:55 UTC; 50ms ago
Docs: man:httpd.service(8)
Main PID: 4050 (httpd)
Status: "Started, listening on: port 80"
Tasks: 177 (limit: 1067)
Memory: 13.2M
CPU: 69ms
CGroup: /system.slice/httpd.service
└─4050 /usr/sbin/httpd -DFOREGROUND
└─4155 /usr/sbin/httpd -DFOREGROUND
└─4156 /usr/sbin/httpd -DFOREGROUND
└─4163 /usr/sbin/httpd -DFOREGROUND
└─4164 /usr/sbin/httpd -DFOREGROUND

Dec 19 10:10:54 ip-10-0-128-113.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Dec 19 10:10:55 ip-10-0-128-113.ec2.internal httpd[4050]: Server configured, listening on: port 80
Dec 19 10:10:55 ip-10-0-128-113.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
[ec2-user@ip-10-0-128-113 ~]$
```

6. Create target groups

- EC2 → Target Groups → Create
- Target type: **Instances**
- Protocol: HTTP
- Port: 80
- VPC: Select your VPC
- Health check path: /
- Click **Create**

EC2 > Target groups

Target groups (1/1) [info](#) [What's new?](#)

Filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
Task	arn:aws:elasticloadbalancing:us-east-1:989615686095:targetgroup/Task/9f52a44282a2279	80	HTTP	Instance	Task4ALB	vpc-0d5307ee979a5df5e

Target group: Task

Details

arn:aws:elasticloadbalancing:us-east-1:989615686095:targetgroup/Task/9f52a44282a2279

Target type
Instance

Protocol : Port
HTTP: 80

Protocol version
HTTP1

IP address type
IPv4

Load balancer
[Task4ALB](#)

VPC
[vpc-0d5307ee979a5df5e](#)

4	2	2	0	0	0
Total targets	Healthy	Unhealthy	Unused	Initial	Draining
	0 Anomalous				

► Distribution of targets by Availability Zone (AZ)

Color values in this table use [resourcecolor](#) filter applied to the [Distribution of targets](#) table below

7. Create Application Load Balancer

- EC2 → Load Balancers → Create
- Choose **Application Load Balancer**
- Name: *Task-APB*
- Scheme: **Internet-facing**
- IP type: IPv4
- Listeners: HTTP 80
- Subnets:
- Select **2 public subnets (different AZs)**
- Security group: ALB-SG
- Listener action:
- Forward to **Target Group**
- Create Load Balancer

EC2 > Load balancers > Create Application Load Balancer

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [info](#)

Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves Internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and **Dualstack** IP address types.

Load balancer IP address type [info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**

Includes only IPv4 addresses.

☐ **Dualstack**

Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with Internet-facing load balancers only.

Network mapping [info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0e5dd4080ca8877c (Task4-vpc)

10.0.0.0/16

[Create VPC](#)

EC2 > Load balancers > Create Application Load Balancer

Availability Zones and subnets [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az1)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-03bce0811c7e59591 IPv4 subnet CIDR: 10.0.0.0/20 Task4--subnet-public1-us-east-1a

☒ **us-east-1b (use1-az2)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
subnet-0edbd8c150446b7ce IPv4 subnet CIDR: 10.0.16.0/20 Task4--subnet-public2-us-east-1b

Security groups [Info](#)
A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups
Select up to 5 security groups

ALB-SG
sg-0a18bb2bae38e9d08 VPC: vpc-0e5dd40800ca8877c

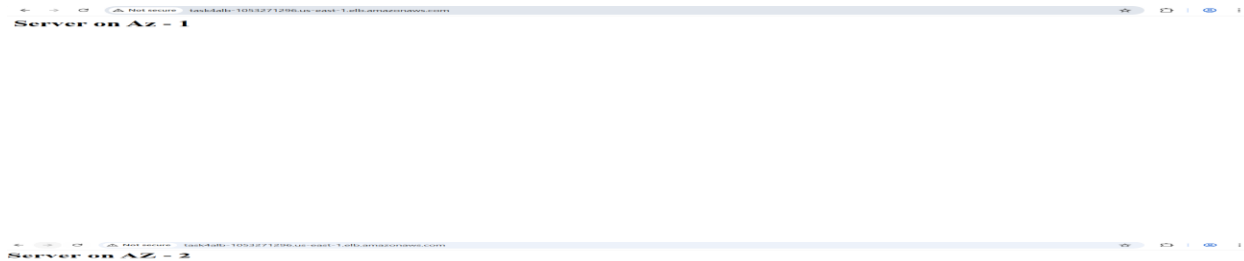
8. Check your Load balancer working through a websites

Goto Load Balancer

Select your Application Load Balancer

Pick DNS name provided by ALB

paste into web browser (<http://>)



9. Create a launch Template helps to launch server

Goto – EC2- Launch Template – Create Template

EC2 > Launch templates > Modify template (Create new version)

▼ Instance type [info](#) | [Get advice](#)

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

☒ All generations
[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

Titoo Template value ▼ [Create new key pair](#)

▼ Network settings [info](#)

Subnet [info](#)

Don't include in launch template ▼ [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone [info](#)

Don't include in launch template ▼ [Enable additional zones](#)

Firewall (security groups) [info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group ☐ Create security group

Common security groups [info](#)

Select security groups ▼

ALB-SG sg-0a99d9b5d1916ba17 [×](#)

[Compare security group rules](#)

Advanced network configuration

▼ Summary

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-068c0051b15c0b816

Virtual server type (instance type)

t2.micro

Firewall (security group)

ALB-SG

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

[Cancel](#) [Create template version](#)

EC2 > Launch templates > Modify template (Create new version)

▼ Network settings [info](#)

Subnet [info](#)

Don't include in launch template ▼ [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone [info](#)

Don't include in launch template ▼ [Enable additional zones](#)

Firewall (security groups) [info](#)

A security group is a set of firewall rules that control the traffic for your instance.

☒ Select existing security group ☐ Create security group

Common security groups [info](#)

Select security groups ▼

ALB-SG sg-0a99d9b5d1916ba17 [×](#)

[Compare security group rules](#)

Advanced network configuration

▼ Storage (volumes) [info](#)

EBS Volumes [Hide details](#)

Volume 1 (AMI Root) : 8 GiB, EBS, General purpose SSD (gp3), 3000 IOPS
AMI Volumes are not included in the template unless modified

Free tier eligible customers can get up to 30 GiB of EBS General Purpose (SSD) or Magnetic storage [×](#)

[Add new volume](#)

▼ Summary

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-068c0051b15c0b816

Virtual server type (instance type)

t2.micro

Firewall (security group)

ALB-SG

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

[Cancel](#) [Create template version](#)

10. Create Auto Scaling Group

EC2- Auto Scaling groups- Create

EC2 > Auto Scaling groups > Create Auto Scaling group

Instance maintenance policy

Replacement behavior

No policy

Min healthy percentage

-

Max healthy percentage

-

Additional settings

Instance scale-in protection

Disabled

Monitoring

Disabled

Default instance warmup

Disabled

Capacity Reservation preference

Preference

Default

Capacity Reservation IDs

-

Resource Groups

-

Step 5: Add notifications

Edit

Notifications

No notifications

Step 6: Add tags

Edit

Tags (0)

Key

Value

Tag new instances

No tags

Preview code

Cancel

Previous

Create Auto Scaling group

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: **Configure group size and scaling**

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Configure group size and scaling - optional

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity

Specify your group size.

1

Scaling

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

1

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy

No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template

Step 2: Choose instance launch options

Step 3 - optional: **Integrate with other services**

Step 4 - optional: Configure group size and scaling

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Integrate with other services - optional

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.

Attach to an existing load balancer

Choose from your existing load balancers.

Attach to a new load balancer

Quickly create a basic load balancer to attach to your Auto Scaling group.

VPC Lattice integration options

To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

No VPC Lattice service

VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.

Attach to VPC Lattice service

Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

Create new VPC Lattice service

Application Recovery Controller (ARC) zonal shift - new

During an Availability Zone impairment, target instance launches towards other healthy Availability Zones.

Enable zonal shift

New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks

Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

CloudShell Feedback Console Mobile App

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template (selected)
 Step 2: Choose instance launch options
 Step 3 - optional: Integrate with other services
 Step 4 - optional: Configure group size and scaling
 Step 5 - optional: Add notifications
 Step 6 - optional: Add tags
 Step 7: Review

Choose launch template info

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name
 Auto Scaling group name
 Enter a name to identify the group.

 Must be unique to this account in the current Region and no more than 255 characters.

Launch template info
 For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
 Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Create a launch template

Version
 Create a launch template version

Description Apache	Launch template Apache lt-0415bce12bcd6f21ac	Instance type t2.micro
AMI ID ami-068c0051b15c8b816	Security groups -	Request Spot Instances No
Key pair name Titoo	Security group IDs sg-0cf99d965d1916ba17	

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template (selected)
 Step 2: Choose instance launch options
 Step 3 - optional: Integrate with other services
 Step 4 - optional: Configure group size and scaling
 Step 5 - optional: Add notifications
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Create a launch template

Cancel **Next**

For Checking All Setup

Increase the Stress on instance

sudo apt install stress-ng

stress-ng --cpu 0 --cpu-method all --metrics-brief

Check the Server , New Server will launch as per Existing server crash

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:

EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Instances (5)

Find instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	public2	i-0801c010326d9a8b0	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	ec2-18-212-136-226.co...	18.212.136.226	-	-
☐		i-0d325091f5a2e2b8b	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	-	-	-	-
☐	private1	i-0c39b63cf2350e5ce	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	-	-	-	-
☐		i-0d275f49c7149c049	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-	-	-	-
☐	private2	i-045b49d43b4b28617	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	-	-	-	-

Select an instance

CloudShell

Feedback

Console Mobile App

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